Mandatory Assignment 3 - Mathematic Methods

Due 5 Apr by 23:59 **Points** 100 **Submitting** an external tool **Available** 19 Jan at 0:00 - 5 Jun at 23:59

Note: Please start with the template provided in Codegrade. You only need to add things and not delete anything from the given template. Otherwise, Codegrade will throw many errors. The main method is already given to you, your task is to write all the other missing methods

LEARNING OBJECTIVES

The student must show ability and skill in creating and using methods in their program logic to encapsulate code in reusable sections to create more extensive programs. Also learn more about tokenization using Scanner class' next() and nextInt() methods.

The most typical mistakes that students do:

- they do not read the instructions carefully. Please see how you will be graded in the section
- they do not use constants
- they do not read the coding rules
- they do the assignment without reading about the theory
- they have not seen the assignment five overview video on Canvas

Prerequisites

The assignment requires you to study the material in module 4 and bring your knowledge and feedback from previous assignments.

Task

This task involves creating several methods that perform various mathematical tasks and calling these from the main method.

Begin the task by constructing a flow chart and program logic on paper. Then, describe your structure in text form (pseudocode) in the comment header of the program so that an outsider can understand what the program does and how your logic/algorithm works.

Methods to implement

The following methods must be implemented

Area of the circle

You must create a method `areaCircle(int radius)` that takes in the radius (positive number) and calculates the base area, in other words, the area of the circle.

The area is returned from the method as a floating point number. Note that you can use float and double for floating point numbers in Java. In this case only one of them is correct, you should know which one to use. Hint: You will be using Math.sqrt() and Math.pow(). What type will they return?

The base area is calculated as

$$B = \pi r^2$$

Hint: To calculate the power (e.g., radius r raised to exponent 2), use the library java method `Math.pow(double a, double b)` where a is the radius and b is the exponent. For PI, you can use the constant `Math.PI`

Lateral surface area of the cone

A cone is a geometric figure with a base surface and a cone surface formed into a tip from the base surface. If the base surface has the shape of a circle and the tip of the cone is directly above or below the center of the circle, then we call the cone a straight, circular cone.

You must create a method `areaCone(int radius, final int height)` that takes in the radius and height and calculates the area of the cone surface.

The area is returned from the method as a floating point number.

The area of the cone surface is calculated as

$$M = \pi \cdot r \cdot s$$

(how to obtain `s` is described in the next section)

Note that you will have two methods called `area`. Having two (or more) methods with the same name is called overloading, and it is the number and type of input parameters that determine which of the methods is run. It is a requirement to have overloaded methods.

Pythagoras' theorem

To calculate the area of the surface of the cone, we must first obtain `s` (i.e., the hypotenuse), and for that, we use Pythagoras' theorem:

$$a^2 + b^2 = c^2$$

You should create a method `pythagoras(int side, final int side)` that takes in a and b (positive numbers) and returns the hypotenuse c as a floating point number. Note that you can use float and double for floating point numbers in Java. In this case only one of them is correct, you should know which one to use. Hint: You will be using Math.sqrt() and Math.pow(). What type will they return?

To get the square root of a number, use the library Java method `Math.sqrt(double a)`.

Volume

You should create a method `volume(final int radius, final int height)` that takes in a radius and height (positive numbers) and calculates the volume.

The volume is returned from the method as a floating point number. Note that you can use float and double for floating point numbers in Java. In this case only one of them is correct, you should know which one to use. Hint: You will be using Math.pow(). What type will they return?

Calculation of the cone's volume is determined using the following formula:

$$V = \frac{\pi r^2 h}{3}$$

Fractions

You should create a method `Fraction(final int numerator, final int denominator)` that takes in a numerator and a denominator and converts these into an integer part and a fraction, and shortens the fraction part as far as possible. For example, when the numerator is 32, and the denominator is 12, you should get 2 2/3, not 2 8/12.

The result must be returned in an integer array, int[3], where the integer part is in first place, and the fraction's numerator and denominator are in second and third place in the array. For example, numerator 7 and denominator 3 should give the result 2 1/3, so the method returns an array that contains [2, 1, 3].

If the denominator entered is 0, the division should not be performed, and the method should instead return null (since division with 0 is not allowed). If the numerator submitted is 0, the method must return [0, 0, 0]

Pseudocode for the fraction (example)

```
1. Make sure that a i s 0 (change the place if necessary)
2. If the denominator is 0 then
   2.1 r = 0
   2.2 w = 0
   2.3 n = 0
3. Return a
```

Printing

You should create a `printFraction(final int[] parts)` method that takes an integer array and presents it as `integer numerator / denominator`.

If the first place in the array is 0, there is no integer part. If the second place in the array is 0, then there is no fraction part; if the last place in the array is also 0, the printout should be 0.

If the array is null, the text "Error" should be printed.

The table below shows examples of what to print depending on what is in the array:

Array	Print
[2, 1, 3]	2 1/3
[0, 2, 5]	2/5
[3, 0, 7]	3
[0, 0, 0]	0
null	Error

Input

You should create a method `input()` that reads the user's input and returns either a positive integer (which means the user entered a valid value >= 0) or -1. Entering q, the method returns -1. If the user enters a negative number, this number should be converted to a positive number by taking the absolute of the number `x = Math.abs(x);` Please note that if there is some other input such as 0.9 not unsatisfactorial input, then the method just ignores that input and reads the next valid value. In other words, everytime input() is called, it must either return -1 to say q was entered or return valid integer, else it just stays in the method until a valid value is entered.

Testing of methods

The main program, i.e., the main method, must test the methods, i.e., call methods with input parameters that are retrieved from the user's input data. The main method is already given to you

Your program should repeatedly read a radius and height (in that order) and test the area and volume methods until the user enters 'q'. Then the program should continue, but now it should read in numerator and denominator (in that order) and test the fraction part. This is repeated until the user enters 'q' again when the program should end.

Appropriate prints are made in the program to be able to follow how the program works. Return values from the area and volume methods are printed by the main method, rounded to two decimal places when printing. The fraction of the fraction method is printed via the printFraction method. Note that it is only the result of the fraction part, which is printed via printFraction, i.e., the answer you get back from the method fraction. A printout of the numerator and denominator you enter, and the equal sign that precedes the answer in the example below is thus printed directly in the main.

Error checks

All inputs in the program are integers except `q`, which ends the program. The input of anything other than integers or `q` should be ignored. The program should not crash regardless of what the user enters (e.g., if, for example, you happen to enter a letter or a floating point number instead of an integer).

The reading should be repeated if the user does not type a `q`. Entering 'q' should cause the program to skip to the next part (fraction part) (or end if q was entered in the fraction part).

Example run of the program

Below you see an example of how the program should work. The example is designed so the program can be quickly and easily tested for correction, so follow it as closely as possible.

```
# Test of area and volume methods

r 2 h 5 A = 31 V = 41
r = 2 h = 4
Cone area: 52.57
Cone area: 20.10
Cone Volume: 16.76

# Test of fraction
r = 2 h = 4 A = 20.10
Cone area: 54.96
Cone Volume: 67.12

# Test of the fraction2 methods

r 7 h 12 B 6 test 3 string -9 B 5 -7 q
r = 7 h = 12
b12 = 6
test = 3 string
b7 = b7=7
```

It should, therefore, be possible to enter several values before pressing enter (after the entered input is sent to the Scanner object). As usual, you use the Scanner methods hasNext,... to check the data type and next,... to load one value at a time from the input buffer (until the next separator). The remaining values remain in the input buffer and wait for their turn to be read.

So, as you know, this only works if you create the Scanner object only once in the program. So it is best to declare the Scanner object in the class body instead of in the method so that the same Scanner object can be used for all inputs.

NOTE! The scanner variable is the only variable that can be declared in the class body.

How will you be graded?

- You will be graded and checked for functional and non-functional requirements
- Functional requirements
  - Integers can be entered and processed until there is a 'q' read. This is done twice
    - First, the area of a circle, the area of a cone, and the volume of the cone calculations and printing
    - Second for calculating the fraction and returning the array
  - The program should never crash and never terminate. When an invalid entry is made, the program should skip or wait for the user to enter a new value.
  - Individual method requirements, as described in each method description above, are also followed.
- Non Functional requirements
  - Strictly follow the coding rules

Codegrade specific Instructions

- Coding rules and styles are automatically checked by the tool
- Make sure to use the template provided on codegrade
- The main method is already given to you. You need to add the missing methods only.
- Do not use nextLine() anywhere in the code, do not use split() method of string either. This is the most common mistake many do.

Submit the solution

Before you turn in the assignment, please ensure your code follows the course's formatting and coding rules requirements. Codegrade does this check and a few other things; make sure you get green tick marks on Codegrade before you turn in the assignment. CodeStyle Blue Info messages are OK. However, all yellow and red Errors must be taken care of.

Plagiarism

When writing the code, make the code your own. Submissions may be checked for plagiarism, and the student will be reported to the university's Disciplinary Board in case of cheating.

VIDEO

Video from the session on 2025-10-27 - Step by Step plan

Quick overview from 2025-03-26 - Must watch

How to implement a given pseudocode into Java code (video from previous year)

Videos from the previous year (Note: the parameters are not final in the below videos, you will need to do slight changes to get it working)

0:00/0:00

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This tool needs to be loaded in a new browser window

Load Mandatory Assignment 3 - Mathematic Methods in a new window